

100% NPU-Native Deployment on Qualcomm Dragonwing IQ-9075 EVK

From CPU Loops to Static Tensor Graphs

AI Hub Verification: QNN DLC, HTP v73, Compute Unit NPU

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Abstract

We prove that four CPU-bound pipeline stages (BPE tokenization, duration expansion with `repeat_interleave`, sliding-window chunking, INT8 dequantize and trimming) can be rewritten as static tensor operations natively supported by the Qualcomm Hexagon Tensor Processor (HTP) on Dragonwing IQ-9075 EVK. All test graphs compile with Qualcomm AI Hub Workbench to QNN DLC and profile **100% NPU**, **0 fallback** on real EVK hardware. We report compile/profile/inference job IDs, latency and peak memory.

1 Hardware and Toolchain

SoC QCS9075 / IQ-9075: Dual Hexagon Tensor Processor, quad HVX + dual HMX per HTP, 8 MB vector-TCM + 1 MB L2 per HTP, up to 100 dense TOPS / 200 sparse TOPS. Software: Qualcomm AI Runtime (QAIRT) 2.45.0, QNN HTP backend v73, target runtime `qnn_dlc`, `-compute_unit npu`. Reference device: **Dragonwing IQ-9075 EVK** on Qualcomm AI Hub. HTP whitelist used: Gather, MatMul, Add/Sub/Mul/Div, Cast, CumSum, GreaterEqual/Less, And (BOOL8), Softmax, LayerNorm, Gelu, Reshape, Transpose, Split, Slice (StridedSlice), Conv2d, FullyConnected, ArgMax, Pad.

2 Problem 1: Text Tokenizer and LLM

Challenge. BPE, UTF-8 decode and dictionary lookup need dynamic branching (`if/while`, hash table). QNN HTP has no `If` / `Shape` / string type, and `Conv1d=NO` on all precisions.

Native solution. Stop treating text as strings. Treat it as a fixed-shape integer tensor $B \in \{0, \dots, 255\}^{1 \times N}$, $N = 512$. Replace BPE by

$$E = \text{Gather}(W_{\text{byte}}[256, D], B) \in \mathbb{R}^{1 \times N \times D}, \quad (1)$$

i.e. `nn.Embedding(256, D)`, followed by a byte-level Transformer using only MatMul, Softmax, LayerNorm, Gelu, Reshape, Transpose, FullyConnected. No OneHot+MatMul (wasteful, dtype mismatch since MatMul OTHERS=NO); no Conv1d (rewrite to Conv2d with $H = 1$).

Verification. Tiny $D=128, L=2$: compile `jp1n6do8g` **SUCCESS**, profile `jpg9zd32wp` **SUCCESS**, infer `j57e9jnvp` **SUCCESS**, **72/72 NPU**, 0.694 ms, 4.64 MB. Big $D=512, L=8$, 25.45M params: compile `j5w14djmp` **SUCCESS**, profile `jpyom4915` **SUCCESS**, infer `jp0m61nng` **SUCCESS**, **256/256 NPU**, 8.505 ms, 4.67 MB. Example $H_i = [228, 189, 160, 229, 165, 189, 72, 105] + \text{pad} \rightarrow \text{logits } [1, 1, 512, 256]$ on device.

Weights. Retrain/distill required (30M–150M byte model, e.g. ByT5-Small). Difficulty 10/10 for quality parity.

3 Problem 2: Duration Expansion

Challenge. `repeat_interleave` copies phoneme frames by dynamic counts on CPU.

Native solution. Build a binary stencil $M \in \{0, 1\}^{1 \times T \times P}$ then stamp:

$$e = \text{cumsum}(d), \quad s = e - d, \quad (2)$$

$$M_{t,p} = \mathbb{K}[t \geq s_p] \wedge \mathbb{K}[t < e_p], \quad (3)$$

$$Y = M \times H, \quad Y \in \mathbb{R}^{1 \times T \times D}. \quad (4)$$

Ops: CumSum, GreaterEqual, Less, And, Cast, MatMul, Sub. All in HTP whitelist. Tested $P=32, T=96, D=64$.

Verification. Compile `jgzlnv345` SUCCESS, profile `jgzlnv2z5` SUCCESS, **12/12 NPU**, 0.254 ms, 6.16 MB. Correctness: $Y[t] = H[t//3]$ when $d_p = 3$. **0% retrain. Difficulty 3/10.**

4 Problem 3: Sliding-Window Chunking

Challenge. CPU loop slices audio into 64-frame blocks.

Native solution. (A) Streaming: fix graph input to $[1, 192, 64]$, let MCU/DMA accumulate 64 frames and interrupt NPU. (B) In-NPU batching: $[1, 1536, 64] \rightarrow \text{Reshape} \rightarrow [8, 192, 64]$ to process 8 chunks in parallel. Overlapping windows via Conv2d with stride $<$ kernel (Unfold equivalent).

Verification. Graph with Reshape + Conv2d + Add + Mul: compile `jgdd27xkg` SUCCESS, profile `j5wl4mwzp` SUCCESS, **8/8 NPU**, 0.555 ms, 7.41 MB. **0% retrain. Difficulty 1/10.**

5 Problem 4: Dequantize and Artifact Trimming

Challenge. INT8 $\rightarrow$ FP16 cast and cutting head/tail margins.

Native solution. Dequantize statically:

$$y = (x - z) \cdot s, \quad \text{via Sub} \rightarrow \text{Mul} \rightarrow \text{Cast}, \quad (5)$$

(QNN also auto-inserts Quantize/Dequantize at graph borders). Trimming: static Slice $x[:, 16:176, :]$ (StridedSlice) or shape-preserving Tukey mask $y = t \odot m$, $m = 0$ at margins, 1 center, via elementwise Mul.

Verification. Graph Cast+Sub+Mul+Slice: compile `jgnzkr0vg` SUCCESS, profile `jg9zd90qp` SUCCESS, **9/9 NPU**, 0.163 ms, 6.32 MB. **0% retrain. Difficulty 2/10.**

6 Summary Results on IQ-9075 EVK

Task	Method	Jobs (C/P/I)	NPU	Lat (ms)	Peak (MB)
1 Tokenizer tiny	Gather Emb.	<code>jp1n6do8g/jg9zd32wp/j57e9jnvp</code>	72/72	0.69	4.6
1 Tokenizer big	Gather Emb.	<code>j5wl4djmp/jpyom49l5/jp0m61nng</code>	256/256	8.51	4.6
2 Duration	CumSum $\rightarrow$ Mask $\rightarrow$ MatMul	<code>jgzlnv345/jgzlnv2z5</code>	12/12	0.25	6.1
3 Chunking	Reshape/Conv2d	<code>jgdd27xkg/j5wl4mwzp</code>	8/8	0.56	7.4
4 Deq+Trim	Sub/Mul/Cast/Slice	<code>jgnzkr0vg/jg9zd90qp</code>	9/9	0.16	6.3

All jobs: device `Dragonwing IQ-9075 EVK`, `-target_runtime qnn_dlc -compute_unit npu`, QAIPT 2.45.0. Problem 1 trades quality for NPU-purity and needs retraining; problems 2–4 are drop-in, weight-preserving, low difficulty.